Chapter 3
Vector Spaces

3.2 Subspace, null space, column space and row space

Learning outcomes

- Be able to show whether a subset is a subspace or not
  判断向量空间的子集是不是“子空间”，要检查加法和标量乘法是否闭合
- Be familiar with examples of subspaces:
  - Null space of a matrix “齐次方程组的解集”
  - Span of vectors “所有线性组合”
  - Column space “列向量的所有线性组合”
  - Row space “行向量的线性组合”
- Be able to check whether a set of vectors is a spanning set of $\mathbb{R}^{n}$, by row reduction, or by finding a vector not in the span.

Subspaces of Vector Spaces

Definition

Let V be a vector space (for example, $\mathbb{R}^n$). A nonempty subset S of V is a **subspace** if it is closed under addition and closed under scalar multiplication.

In other words,

(1) $\mathbf{u}, \mathbf{v} \in S \Rightarrow \mathbf{u}+\mathbf{v} \in S$ (*S* is closed under addition)

(2) $\mathbf{u} \in S, c \in \mathbf{R} \Rightarrow c\mathbf{u} \in S$ (*S* is closed under scalar multiplication)

To show that a subset S of a vector space forms a subspace, we must show that

1. S is nonempty (it contains the zero vector)
2. The closure properties (i) and (ii) are satisfied.

S is a subspace of a vector space V if it is

① a non-empty subset of V

$S \neq \emptyset$ or $\exists\, x \in S$.

$S \subseteq V$

② closed under addition

$\forall\, x, y \in S \quad x + y \in S$

③ closed under scalar multiplication

$\forall\, \alpha \in \mathbb{R} \quad \forall\, x \in S \quad \alpha x \in S$

① $S \subseteq V$

$\forall\, x \in S, \; x \in V$

② $V_1 = V_2$

$V_1 \subseteq V_2 \;\&\; V_2 \subseteq V_1$

Properties of a subspace S of V.

① S must contain the "0" and "$-x$" and the "0" and "$-x$" are the same in V.

Since S is a subspace of V

→ { ⓐ it is closed under scalar multiplication
  ⓑ $S \subseteq V$

ⓐ → $\forall \alpha \in \mathbb{R}$ $\forall x \in S$, $\alpha x \in S$

If $\alpha = 0$ $0x \in S$

If $\alpha = -1$ $(-1)x \in S$

ⓑ Since $S \subseteq V$, $\forall x \in S$, $x \in V$

Since V is a vector space.

$0 \cdot x = "0" \in V$ $(-1) \cdot x = "-x" \in V$

$"0" = 0 \cdot x \in S$ $"-x" = (-1) \cdot x \in S$

⇒ S and V share the same "0".

If S is a subspace of V.

then "0" in V is also in S.

↓

If "0" in V is not in S

S is not a subspace of V.

If $p \to q$

If not $q \to$ not p

To check S is a subspace or not

we firstly check "0" in V $\in S$.

if yes. S is non-empty

if not. S is not a subspace

since it is not closed

under scalar multiplication

($\alpha = 0$ $\quad 0 \cdot x \notin S$).

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(2). A subspace itself is a vector space.

1> Since $S \subseteq V$.

$\Downarrow$

$\forall x \in S. \quad x \in V$.

Since V is a vector space. any vector in V satisfies A1-A8.

2> S satisfies C1 & C2

$\Rightarrow$ S is a vector space.

Example 1

Consider the subset W of $\mathbb{R}^2$ of vectors of the form $[a, 2a]^T$
Show that W is a subspace of $\mathbb{R}^2$.

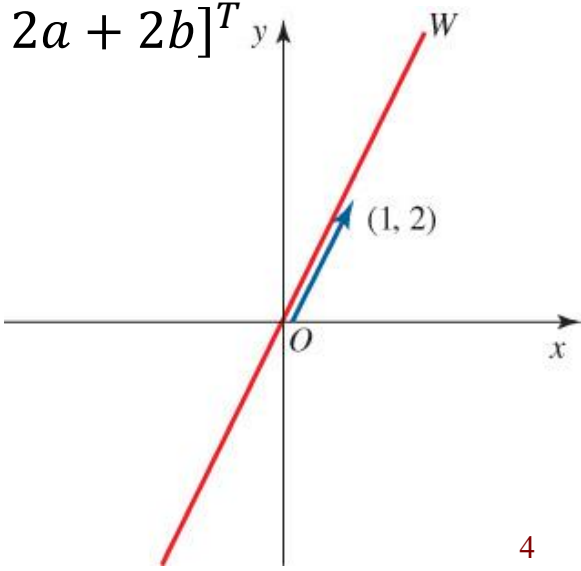
Proof Observe: W is the set of vectors that can be written as $a[1,2]^T$

- W is non-empty because $[0,0]^T \in W$

Let **u** = $[a, 2a]^T$, **v** = $[b, 2b]^T \in W$, and $k \in \mathbf{R}$.

- **u** + **v** = $[a, 2a]^T + [b, 2b]^T = [a + b, 2a + 2b]^T$
  $= [a + b, 2(a + b)]^T \in W$,
- $k\mathbf{u} = k[a, 2a]^T = [ka, 2ka]^T \in W$.

Thus **u** + **v** $\in W$ and $k\mathbf{u} \in W$.
W is closed under addition and scalar multiplication.
Therefore W is a subspace of $\mathbb{R}^2$.



$$W = \{ a[1, 2]^T \mid \forall a \in R \}$$

① $W \subseteq R^2$.

$[0, 0]^T \in W$

$\}$ → W is a non-empty subset of R^2.

② $\forall x = [a, 2a]^T \in W \quad \forall y = [b, 2b]^T \in W$

$x + y = [a+b, 2(a+b)]^T \in W$

③ $\forall \alpha \in R \quad \forall x = [a, 2a]^T \in W$

$\alpha x = [\alpha a, 2(\alpha a)]^T \in W$

$\Rightarrow$ W is a subspace of R^2.

Example 2

Consider the homogeneous system of linear equations. It can be shown that there are many solutions $x_1=2r$, $x_2=5r$, $x_3=r$. We can write these solutions as vectors in $\mathbb{R}^3$ as $(2r, 5r, r)^T$.

$$\begin{aligned} x_1 - x_2 + 3x_3 &= 0 \\ x_2 - 5x_3 &= 0 \\ 2x_1 - x_2 + x_3 &= 0 \end{aligned}$$

Show that the set of solutions W is a subspace of $\mathbb{R}^3$.

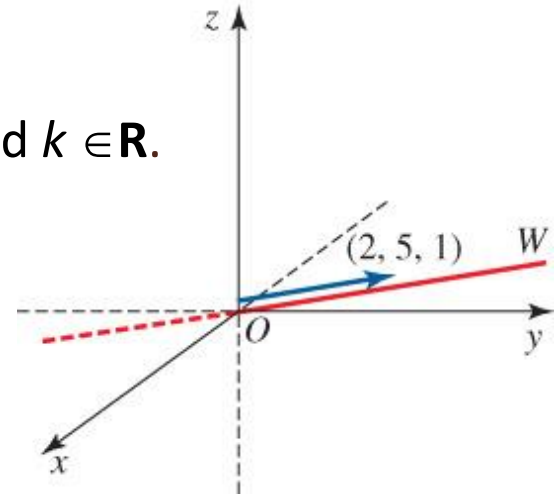
Proof

- W is nonempty: $[0,0,0]^T \in W$

Let $\mathbf{u} = [2r,5r,r]^T$, $\mathbf{v} = [2s,5s,s]^T \in W$, and $k \in \mathbf{R}$.

- $\mathbf{u} + \mathbf{v} = [2(r+s), 5(r+s), r+s]^T \in W$
- $k\mathbf{u} = [2kr, 5kr, kr]^T \in W$

Thus $\mathbf{u} + \mathbf{v} \in W$ and $k\mathbf{u} \in W$,
and W is a subspace of $\mathbb{R}^3$.



Observe: W is the set of vectors that can be written $r(2,5,1)$.

$W = \{ r(2, 1, 1)^T \mid \forall r \in \mathbb{R} \}$

a line in 3-D passing the origin

① $W \subseteq \mathbb{R}^3$ $(0, 0, 0)^T \in W$

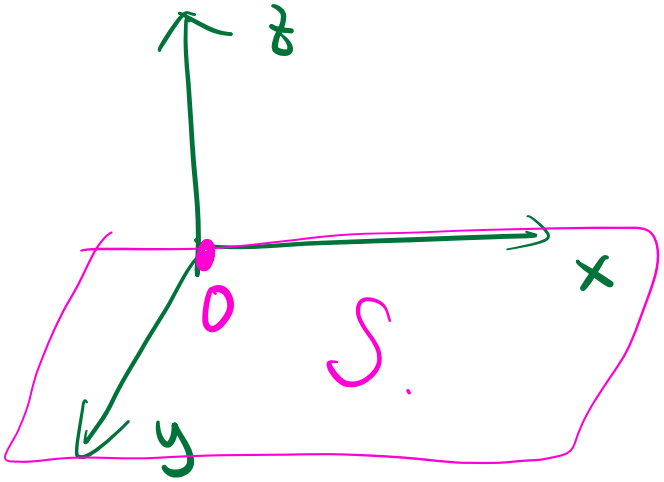
Q: Is $\mathbb{R}^2$ a subspace of $\mathbb{R}^3$

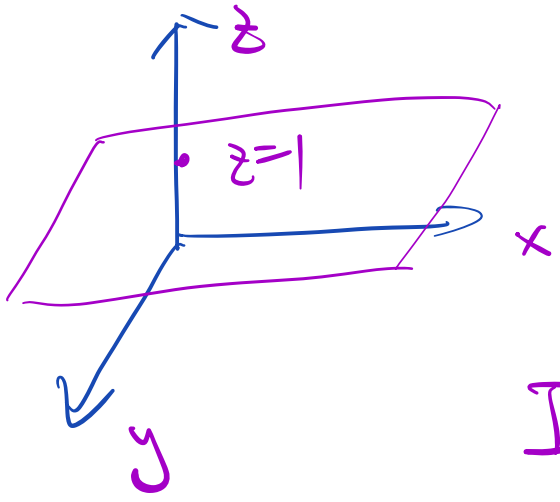
No

$\mathbb{R}^2$ is not a subset of $\mathbb{R}^3$

$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \in \mathbb{R}^2$ $\begin{bmatrix} 1 \\ 2 \end{bmatrix} \notin \mathbb{R}^3$

$S = \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \mid \forall x, y \in \mathbb{R} \right\}$ S is a subspace of $\mathbb{R}^3$





$$W = \left\{ \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \in \mathbb{R}^3 \mid \forall x, y \in \mathbb{R} \right\}$$

Is W a subspace of $\mathbb{R}^3$?

No

$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \notin W$.

W is not closed under scalar multiplication

Since $\forall \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \in W$.

$$0 \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \notin W$$

Exercise

Let $S = \{A \in \mathbb{R}^{2\times 2} : a_{11} = 2a_{22}, a_{21} = 0\}$ with standard addition and scalar multiplication. Prove that S is a subspace of $\mathbb{R}^{2\times 2}$.

Poof: Step 1: $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in S$, therefore S is nonempty. $S \subseteq R^{2\times 2}$

Step 2: Let $A = \begin{bmatrix} 2a & * \\ 0 & a \end{bmatrix}$ and $B = \begin{bmatrix} 2b & * \\ 0 & b \end{bmatrix}$ be any 2 elements in S.

$$A + B = \begin{bmatrix} 2(a+b) & * \\ 0 & a+b \end{bmatrix} \in S$$

So S is closed under addition.

For any $k \in R$,

$$kA = \begin{bmatrix} 2ka & * \\ 0 & ka \end{bmatrix} \in S$$

Therefore S is closed under scalar multiplication.

Therefore S is a subspace of $R^{2\times 2}$.

$$S = \{ A \in \mathbb{R}^{2\times 2} \mid a_{11} = 2a_{22},\ a_{21} = 0 \}$$

$$= \left\{ \begin{bmatrix} 2a & b \\ 0 & a \end{bmatrix} \mid \forall\, a, b \in \mathbb{R} \right\}$$

Trivial subspace

If V is a vector space, then V is a subspace of itself, and S = {**0**} is a **subspace** of V.

All the other subspaces are referred to as ***proper subspaces***.

We refer to {**0**} as the ***zero subspace***.

① V is a subspace of V.

1> $0 \in V$ $V \subseteq V$

2> V is a vector space. it satisfies C1 & C2.

② $S = \{0\}$ is a subspace of V.

① 1> $S \subseteq V$ since $\forall 0 \in S$. $0 \in V$

2> $0 \in S$

(2) $\forall x, y \in S$. $x=0$, $y=0$

$x+y = 0+0 = 0 \in S$

(3) $\forall \alpha \in \mathbb{R}$ $\forall x \in S$ $x=0$

$\alpha x = \boxed{\alpha 0 = 0} \in S$

$\alpha x + \alpha y = \alpha(x+y)$

$\alpha x + \beta x = (\alpha+\beta)x$

$\alpha 0 + \alpha 0 = \alpha(0+0) = \alpha 0$

$= (\alpha+\alpha)0 = 2\alpha 0$

$\Rightarrow 2\alpha 0 = \alpha 0 \Rightarrow \alpha 0 = 0$

Null space of a Matrix:

For an $m \times n$ matrix A, let $N(A)$ denote the set of all solutions to the homogeneous system $A\mathbf{x} = 0$, that is

$$N(A) = \{x \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

$N(A)$ is a **subspace** of $\mathbb{R}^n$ because:

- **Closed under addition**:

  if $\boldsymbol{x}, \boldsymbol{y} \in N(A)$, then $A\boldsymbol{x} = \mathbf{0}$, $A\boldsymbol{y} = \mathbf{0} \Rightarrow A(\boldsymbol{x} + \boldsymbol{y}) = \mathbf{0}$, so $x + y \in N(A)$

- **Closed under scalar multiplication:**

  if $\boldsymbol{x} \in N(A)$, c $\in \mathbb{R}$, then $A\boldsymbol{x} = \mathbf{0} \Rightarrow A(c\boldsymbol{x}) = \mathbf{0} \Rightarrow c\boldsymbol{x} \in N(A)$

The subspace $N(A)$ is called the **null space** of the matrix *A*.

How to find the null space of a matrix A?

Very Important!

(1) $\forall x, y \in N(A)$, $Ax = 0$ $Ay = 0$ $A: m \times n$

(2) $N(A)$ is a subspace of R^n.

(1) $N(A) \subseteq R^n$ $\quad 0 \in N(A)$

since $Ax = 0$ has (always) a trivial solution.

(2) $\forall y, z \in N(A)$, $Ay = 0$, $Az = 0$

$A(y+z) = Ay + Az = 0 + 0 = 0$

$(y+z)$ is a solution to $Ax = 0$

$y + z \in N(A)$

(3) $\forall \alpha \in R$ $\forall y \in N(A)$ $Ay = 0$

$A(\alpha y) = \alpha(Ay) = \alpha \cdot 0 = 0$

αy is a solution to $Ax = 0$

$\alpha y \in N(A)$.

$\Rightarrow N(A)$ is a subspace of R^n.

Example: Very important

Find the null space $N(A)$ for the matrix $\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}$.

① Solve $Ax = 0$ using Gaussian elimination.

② Express the solution set.

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & -2 & 1 & 0 \end{array}\right] \xrightarrow{-R_2}$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 0 \end{array}\right] \xrightarrow{R_1 - R_2} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 \end{array}\right]$$

x_1, x_2 are leading variables.

x_3, x_4 are free variables. Let $x_3 = \alpha$ $x_4 = \beta$

$x_1 = \alpha - \beta$ $x_2 = -2\alpha + \beta$ $x_3 = \alpha$ $x_4 = \beta$

$$\left\{ \begin{bmatrix} \alpha - \beta \\ -2\alpha + \beta \\ \alpha \\ \beta \end{bmatrix} \mid \forall \alpha, \beta \in \mathbb{R} \right\}$$

$$= \left\{ \begin{bmatrix} \alpha \\ -2\alpha \\ \alpha \\ 0 \end{bmatrix} + \begin{bmatrix} -\beta \\ \beta \\ 0 \\ \beta \end{bmatrix} \mid \forall \alpha, \beta \in \mathbb{R} \right\}$$

$$= \left\{ \alpha \underbrace{\begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}}_{v_1} + \beta \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}}_{v_2} \mid \forall \alpha, \beta \in \mathbb{R} \right\}$$

$$= N(A)$$

$$= \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Example: Very important

Find the null space $N(A)$ for the matrix $\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}$.

Solution

Use Goss-Jordan reduction to solve $A\mathbf{x} = 0$, we obtain

$$[\boldsymbol{A}|\mathbf{0}] \sim \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 \end{array}\right]$$

How to determine leading variables and free variables?

The reduced echelon form involves two free variables, x_3, x_4:

$$x_1 = x_3 - x_4$$
$$x_2 = -2x_3 + x_4$$

Write leading variables in terms of free variables

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_3 - x_4 \\ -2x_3 + x_4 \\ x_3 + 0x_4 \\ 0x_3 + x_4 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Then, $N(A) = \left\{ \alpha \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix} | \alpha, \beta \in R \right\}$

Write all the variables in a column vector, and replace the leading variable by free variables

Split the vector x, in terms of linear combinations of known vectors, with free variables as coefficients.

$2a_1 + 3a_2$

The Span of a Set of Vectors

- Definition: For vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ from a vector space V and scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$, the sum $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n$ is called a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$. 线性组合
- The set of all possible linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is called the **span** of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, denoted by $Span(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$: 张成（的空间）

$$span(\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n) = \{\alpha_1\boldsymbol{v}_1 + \alpha_2\boldsymbol{v}_2 + \cdots + \alpha_n\boldsymbol{v}_n : \alpha_i \in R\}$$

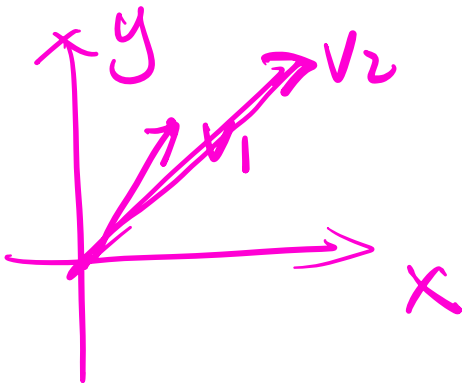
Examples:

- The span of 1 nonzero vector in 2D or 3D is a straight line 直线
- The span of 2 vectors in 3D is (usually) a plane 平面
- In previous example, the null space of A is the span of vectors $[1, -2, 1, 0]^T$ and $[-1, 1, 0, 1]^T$

Eg $v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $v_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$\underline{\text{Span}\{v_1, v_2\}} = \{ 1v_1 + 2v_2,\ -6v_1 + 8v_2,\ 0v_1 + 0v_2, \ldots \}$

$= \{ \alpha_1 v_1 + \alpha_2 v_2 \mid \forall \alpha_1, \alpha_2 \in R \} = R$

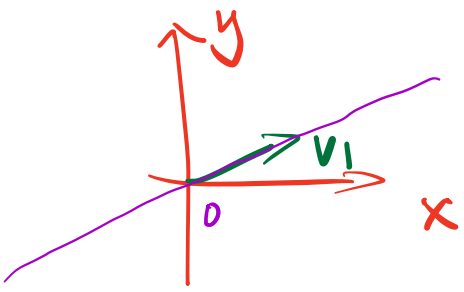


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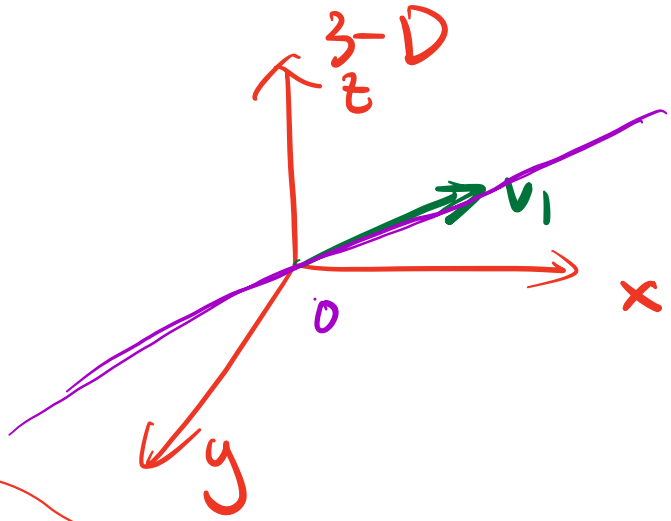
① 1 vector

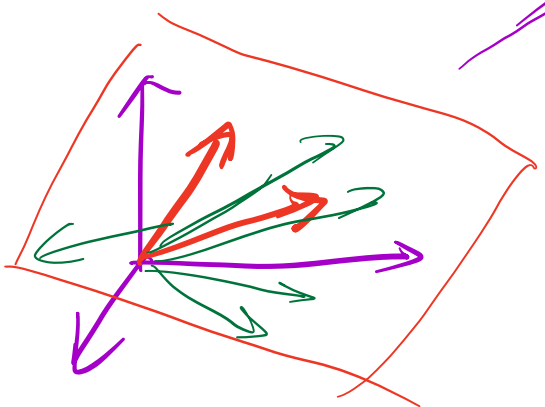
$\text{Span}\{v_1\} = \{ \alpha_1 v_1 \mid \forall \alpha_1 \in R \}$

2-D



3-D





Example:

Given $\mathbf{v} = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$, what is span(**v**)? $= \left\{ \alpha \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} \mid \alpha \in R \right\}$

In words: span(**v**) is the set of vectors of the form

$$\alpha\mathbf{v} = \alpha \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha \\ -2\alpha \\ 0 \end{bmatrix}$$

In notation:

straight line passing through 0 in 3-D.

$$span(\mathbf{v}) = \left\{ \begin{bmatrix} \alpha \\ -2\alpha \\ 0 \end{bmatrix} : \alpha \in R \right\}$$

Theorem

$v_1, v_2, \cdots, v_n \in V$

If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are vectors from a vector space V, then $Span(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ is a subspace of V.

Why is it a subspace?

- If two vectors are both linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, is their sum still a linear combination of the n vectors?
- If a vector is a linear combination, is its scalar multiple still a linear combination?

① Since $0 = 0v_1 + 0v_2 + \cdots + 0v_n \in Span\{v_1 \cdots v_n\}$

$Span\{v_1 \cdots v_n\}$ is non-empty.

Idea $\forall x \in Span\{v_1 \cdots v_n\} \quad x \in V$

$\Rightarrow Span\{v_1 \cdots v_n\} \subseteq V$

$\forall x \in \text{Span}\{v_1 \cdots v_n\}$ $\quad x = \alpha_1 v_1 + \cdots + \alpha_n v_n$

Since $v_1 \cdots v_n \in V$, V is a vector space

then V is closed under addition and scalar multiplication.

$\alpha_1 v_1 \in V \quad \alpha_2 v_2 \in V \quad \cdots \quad \alpha_n v_n \in V$

$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n \in V$

$x \in V$

$\text{Span}\{v_1 \cdots v_n\} \subseteq V$

(2) $\forall x, y \in \text{Span}\{v_1 \cdots v_n\}$

$x = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$

$y = \beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_n v_n$

$x + y = (\alpha_1 + \beta_1) v_1 + \cdots + (\alpha_n + \beta_n) v_n \in \text{Span}\{v_1 \cdots v_n\}$

(3) $\forall k \in R \quad \forall x \in \text{Span}\{v_1 \cdots v_n\}$

$kx = \alpha_1 (k v_1) + \cdots + \alpha_n (k v_n) \in \text{Span}\{v_1 \cdots v_n\}$

Spanning Set

Definition

The set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a **spanning set** (向量空间的）张集 for a vector space V if and only if every vector in V can be written as a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, that is to say, for any $\mathbf{x} \epsilon V$ there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that

$$\mathbf{x} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_n \mathbf{v}_n.$$

如果某些向量$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$是V的张集，则V中任意向量都可以表示为$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$的线性组合

① Spanning set is a small set

$\{v_1, v_2 \cdots v_n\}$

② If $\{v_1, \cdots, v_n\}$ is a spanning set of V.

$\forall x \in V$ $\exists \alpha_1, \alpha_2, \cdots \alpha_n \in R$

such that $x = \alpha_1 v_1 + \cdots + \alpha_n v_n$

③ $\{v_1 \cdots v_n\}$ is a spanning set of V. ⇔

$v_1, \cdots, v_n$ span V

Eg. $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\{v_1, v_2\}$ is a spanning set of R^2.

$\forall x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in R^2$ $x = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

④ Question: If $\{v_1 \cdots v_n\}$ is a spanning set of V. is $V = \text{Span}\{v_1 \cdots v_n\}$?

No.

Eg. $\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$ is a spanning set

for $V = \left\{\begin{pmatrix}x\\y\\0\end{pmatrix} \mid x, y \in R\right\}$

since. $\begin{pmatrix}x\\y\\0\end{pmatrix} = x\begin{pmatrix}1\\0\\0\end{pmatrix} + y\begin{pmatrix}0\\1\\0\end{pmatrix} + 0\begin{pmatrix}0\\0\\1\end{pmatrix}$

$$\mathrm{Span}\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}\right\} = \left\{\alpha\begin{pmatrix}1\\0\\0\end{pmatrix} + \beta\begin{pmatrix}0\\1\\0\end{pmatrix} + \gamma\begin{pmatrix}0\\0\\1\end{pmatrix} \mid \alpha, \beta, \gamma \in R\right\}$$

$$= \left\{\begin{pmatrix}\alpha\\\beta\\\gamma\end{pmatrix} \mid \alpha, \beta, \gamma \in R\right\} = R^3$$

V is a subspace of $\mathrm{Span}\{\ \}$?

⑤ If $\{v_1 \cdots v_n\}$ is a spanning set of V.

$V \subseteq \mathrm{Span}\{v_1 \cdots v_n\}$

$\forall x \in V \quad \exists \alpha_1 \cdots \alpha_n \in R \quad x = \alpha_1 v_1 + \cdots + \alpha_n v_n$

$\in \mathrm{Span}\{v_1 \cdots v_n\}$

$V \subseteq \mathrm{Span}\{v_1 \cdots v_n\}$

(6) If $v_1 \cdots v_n \in V$

$\text{Span}\{v_1 \cdots v_n\} \subseteq V$

(7) (5) & (6) $\Rightarrow$

$\begin{cases} v_1 \cdots v_n \in V \\ v_1 \cdots v_n \text{ span } V \end{cases}$ & $\Rightarrow V = \text{Span}\{v_1 \cdots v_n\}$

(8) $\{v_1 \cdots v_n\}$ is a spanning set of

$\text{Span}\{v_1 \cdots v_n\}$

(9) How to check whether a set is a spanning set of V?

$\forall b \in V \quad \exists \alpha_1 \cdots \alpha_n \in R$

such that $b = \alpha_1 v_1 + \cdots + \alpha_n v_n$

whether $\alpha_1 \cdots \alpha_n$ always exist?

If $V = R^m$

$\forall b \in V = R^m$

$b = \alpha_1 v_1 + \cdots + \alpha_n v_n$ $\quad\quad$ $\alpha_1 \cdots \alpha_n$ exist?

$= \underbrace{[v_1 \cdots v_n]}_{A} \underbrace{\begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}}_{x}$

A: $m \times n$ $\quad$ x: $n \times 1$

Equivalent to check whether

$Ax = b$ is consistent for $\forall b \in R^m$

always.

If Yes. $v_1 \cdots v_n$ span $V = R^m$

If not $\{v_1 \cdots v_n\}$ is not a spanning set of

How to check if a set spans $\mathbb{R}^n$

Is $\{v_1, v_2, ...\}$ a spanning set of $\mathbb{R}^3$?

To determine whether a set spans R^3, we must determine whether an arbitrary vector $\mathbf{b}=[a, b, c]^T$ in $\mathbb{R}^3$ can be written as a linear combination of the vectors in the set.

- Step 1: Put the column vectors into a matrix, $A=[v_1, v_2, \ldots]$
- Step 2: Augment the matrix by the vector $\mathbf{b}=[a, b, c]^T$
- Step 3: Row reduce the augmented matrix $[A|\ \mathbf{b}]$ to check whether it is consistent for ALL a, b, c
- Step 4: If $[A|\ \mathbf{b}]$ is consistent for ALL a, b, c, then the columns of A, $v_1, v_2, \ldots$, is a spanning set of $\mathbb{R}^3$.

Check consistency only !

Spanning set example 1

- Is $\{e_1, e_2, e_3, (1, 2, 3)^T\}$ a spanning set of $\mathbb{R}^3$?

$e_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$ ← ith

$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ $e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$

$\forall b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in \mathbb{R}^3$

$[A|b] = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & b_1 \\ 0 & 1 & 0 & 2 & b_2 \\ 0 & 0 & 1 & 3 & b_3 \end{array}\right]$ is always consistent.

$\{\ldots\}$ is a spanning set for $\mathbb{R}^3$

Spanning set example 1

- Is $\{e_1, e_2, e_3, (1, 2, 3)^T\}$ a spanning set of $\mathbb{R}^3$?

Solution:

It is easily seen that $[a, b, c]^T$ can be written as

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

So the set is a spanning set of $\mathbb{R}^3$

Spanning set example 2

- Is $\{ [1, 1, 1]^T, [1, 1, 0]^T, [1, 0, 0]^T \}$ a spanning set of $\mathbb{R}^3$?

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$\forall b \in \mathbb{R}^3$

$$|A| \overset{c_1 \leftrightarrow c_3}{=\!=} (-1)\begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = -1 \neq 0$$

A is nonsingular $\Leftrightarrow$ $Ax = b$ has the unique solution for $\forall b \in \mathbb{R}^3$

$\{\quad\}$ is a spanning set.

If $|A|=0$, A is singular.

$\Downarrow$

$Ax=b$ doesn't have the unique solution

↙ ↘

no solution → inconsistent

infinitely many → consistent

$[A|b] \to \to$ r.r.e.f

$Ax=b$ consistent for $\forall b \in \mathbb{R}^m$?

↑ spanning set

Spanning set example 2

- Is $\{ [1, 1, 1]^T, [1, 1, 0]^T, [1, 0, 0]^T \}$ a spanning set of $\mathbb{R}^3$?

Solution:

We must determine whether it is possible to find constants α_1, α_2, and α_3 such that

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

This leads to the system of equations

$$\begin{aligned} \alpha_1 + \alpha_2 + \alpha_3 &= a \\ \alpha_1 + \alpha_2 &= b \\ \alpha_1 &= c \end{aligned}$$

Since the coefficient matrix of the system is nonsingular, the system has a unique solution. (See next page)

Spanning set example 2

In fact we find that

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} c \\ b-c \\ a-b \end{bmatrix}$$

Thus,

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = c \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + (b-c) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + (a-b) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

so the three vectors span R^3.

Spanning set example 3

- Is $\{ [1, 0, 1]^T, [0, 1, 0]^T \}$ a spanning set of $\mathbb{R}^3$?

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad \forall b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in \mathbb{R}^3$$

$$[A|b] = \left[\begin{array}{cc|c} 1 & 0 & b_1 \\ 0 & 1 & b_2 \\ 1 & 0 & b_3 \end{array}\right] \xrightarrow{R_3 - R_1} \left[\begin{array}{cc|c} 1 & 0 & b_1 \\ 0 & 1 & b_2 \\ 0 & 0 & b_3 - b_1 \end{array}\right]$$

The system is inconsistent when $b_3 - b_1 \neq 0$

It is not always consistent for $\forall b \in \mathbb{R}^3$

The set is not a spanning set for $\mathbb{R}^3$.

Spanning set example 3

- Is $\{ [1, 0, 1]^T, [0, 1, 0]^T \}$ a spanning set of $\mathbb{R}^3$?

Solution:

(In this example we can clearly see the span of the 2 vectors, and easily find vectors NOT in their span.)

We should note that linear combinations of $[1, 0, 1]^T$ and $[0, 1, 0]^T$ produce vectors of the form $[\alpha, \beta, \alpha]^T$. Thus any vector $[a, b, c]^T$ in $\mathbb{R}^3$ where $a \neq c$ would not be in the span of these two vectors.

Therefore $\{[1, 0, 1]^T, [0, 1, 0]^T\}$ is NOT a spanning set of $\mathbb{R}^3$.

Spanning set example 4

- Is $\{ (1, 2, 4)^T, (2, 1, 3)^T, (4, -1, 1)^T \}$ a spanning set of $\mathbb{R}^3$?

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & -1 \\ 4 & 3 & 1 \end{bmatrix} \qquad \forall b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in \mathbb{R}^3$$

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 2 & 4 & b_1 \\ 2 & 1 & -1 & b_2 \\ 4 & 3 & 1 & b_3 \end{array}\right] \begin{array}{l} R_2 - 2R_1 \\ \longrightarrow \\ R_3 - 4R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & b_1 \\ 0 & -3 & -9 & b_2 - 2b_1 \\ 0 & -5 & -15 & b_3 - 4b_1 \end{array}\right] \xrightarrow{R_3 - \frac{5}{3}R_2}$$

(Continue to next page)

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & b_1 \\ 0 & -3 & -9 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 - 4b_1 - \frac{5}{3}(b_2 - 2b_1) \end{array}\right]$$

If _____ $\neq 0$, the linear system will be inconsistent.

{ } is not a spanning set

Spanning set example 4

- Is $\{(1, 2, 4)^T, (2, 1, 3)^T, (4, -1, 1)^T\}$ a spanning set of $\mathbb{R}^3$?

Solution:

Similar to Example 2. If $\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + \alpha_3 \begin{bmatrix} 4 \\ -1 \\ 1 \end{bmatrix}$ then

$$\begin{aligned} \alpha_1 + 2\alpha_2 + 4\alpha_3 &= a \\ 2\alpha_1 + \alpha_2 - \alpha_3 &= b \\ 4\alpha_1 + 3\alpha_2 + \alpha_3 &= c \end{aligned}$$

In this case, however, the coefficient matrix is singular. Gaussian elimination will yield a system of the form

$$\begin{aligned} \alpha_1 + 2\alpha_2 + 4\alpha_3 &= a \\ \alpha_2 + 3\alpha_3 &= \frac{2a - b}{3} \\ 0 &= 2a - 3c + 5b \end{aligned}$$

(Continue to next page)

If $2a - 3c + 5b \neq 0$, then the system is inconsistent. Hence, for most choices of *a*, *b*, and *c*, it is impossible to express $[a, b, c]^T$ as a linear combination of $[1, 2, 4]^T$, $[2, 1, 3]^T$, and $[4, -1, 1]^T$. The vectors do not span R^3.

Row space & column space

- For an $m \times n$ matrix A, the rows are n-vectors from $\mathbb{R}^{1\times n}$, and the column vectors are m-vectors from $\mathbb{R}^m$.

Definition

For an $m \times n$ matrix A, the subspace of $\mathbb{R}^{1\times n}$ spanned by the row vectors of A is called the **row space** of A. The subspace of $\mathbb{R}^m$ spanned by the column vectors of A is called the **column space** of A.

矩阵A的行空间：矩阵A所有行向量的张成（是$\mathbb{R}^{1\times n}$的子空间）
矩阵A的列空间：矩阵A所有列向量的张成（是$\mathbb{R}^m$ 的子空间）

Example

- The 3×4 matrix

$$A = \begin{bmatrix} 1 & 1 & 2 & -1 \\ 2 & 0 & -4 & 3 \\ 1 & -3 & -14 & 9 \end{bmatrix}$$

has a row space in $\mathbb{R}^{1\times 4}$ given by $\subseteq \mathbb{R}^{1\times 4}$

$$Span(\,[1,1,2,-1],[2,0,-4,3],[1,-3,-14,9])$$

and a column space in $\mathbb{R}^3$ given by

$$Span\left(\left(\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 2 \\ -4 \\ -14 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 9 \end{bmatrix}\right)\right) \quad \subseteq \mathbb{R}^3$$